

DeepRobotics Jueying Lite3 — How to connect

Audience: operator with the robot and a laptop. Time: ~2 minutes. No third-party software required.

1. Power and network

1. Power on the robot and let it finish booting (about 1 minute).
2. The robot creates its own Wi-Fi hotspot: **YSC-JYML-XXXXX** (your unit's exact name is on its QR label).
3. On the operator laptop, join that hotspot. The password is on the robot's QR label.
4. You should get an address in **192.168.2.x**; the robot's motion host is **192.168.2.1**.
5. Check the link: ping 192.168.2.1 should reply.

Ethernet alternative: with the robot wired to the same router, the motion host is **192.168.1.120**; set that IP where the instructions below say 192.168.2.1. Requiring internet at the same time? Stay on the wired network instead of the hotspot.

2. Connect with the web console (recommended)

1. Start the console: open the **Lite3 Pilot** app (or run `python3 server.py --port 8123` in the app folder).
2. In the browser window that opens, check the header shows the robot IP (default **192.168.2.1**; change it under **Advanced** → **Robot IP** if needed).
3. Press  **Connect**. The status must show  **Connected** and a live ping.
4. The camera opens by itself within ~10 s (if the robot's camera service is running).
5. Motion is enabled once connected: stand/sit, actions and the joysticks.

3. Connect with the Python SDK (scripting)

1. In a terminal: `cd lite3_sdk && python3 -m lite3sdk.cli status --ip 192.168.2.1`
2. Health check: `python3 -m lite3sdk.cli stand` — the robot stands.
3. Drive: `python3 -m lite3sdk.cli drive --vx 0.1 --seconds 2` (slow forward, then stops).
4. In Python: `dog = lite3sdk.Lite3("192.168.2.1"); dog.connect(); dog.stand(); dog.drive(vx=0.1); dog.wait(2); dog.stop()`

4. Safety before moving

- Clear the area (5 m), leash or stand the robot for first tests.
- Keep the physical E-stop in reach — it always overrides software.
- If the software E-stop is latched in the console, press  **Return** to re-arm. It is latched by design: no motion commands are sent until then.
- **The robot must be standing for velocity/driving commands to be honoured.** If it is sitting/lying, motion frames are ignored silently.

5. Troubleshooting

- **Not connected / ping fails:** wrong network (must be the robot hotspot or its wired LAN), wrong IP in Advanced, or the robot is still booting.
- **Commands do nothing:** check the software E-stop (press  Return), confirm the robot is standing, and confirm the console shows  Connected.

- **No camera:** the robot's camera service must be running on the robot; the console retries automatically every ~8 s.
- **No battery / IMU readout:** one-time robot-side setting — in `~/jy_exe/conf/network.toml` set `ip = '<laptop IP>'` and restart the motion program. Motion works without it.
- **UDP gives no acknowledgement** — "command sent" does not prove motion; confirm visually (or with telemetry).

Ports used: commands UDP 43893 · telemetry UDP 43897 · camera RTSP 8554 (/test) · remote/gamepad UDP 12121 · console HTTP 8123.